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Are Both Quantity and Quality of Green Innovation Exalted in Polluting Firms Under Environmental Regulation?

Sin-Som Sergio Tsiong[1] | Jiaqi Feng[2] | Mingqian Zhang[3]

[1]School of International and Public Affairs, Shanghai Jiao Tong University, Shanghai, China | [2]Department of Economics, Korea University, Seoul, Korea | [3]School of Economics and Finance, Shanghai International Studies University, Shanghai, China

Correspondence: Sin-Som Sergio Tsiong (s.sergio.zhang@gmail.com)

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ABSTRACT

Green innovation plays a crucial role in the context of the low-carbon economy. Still, it is equally important to focus on gatekeeping the quality of innovation amidst the plethora of innovative outcomes. This article aims to investigate how China's carbon emission trading (CET) pilot will affect both the quantity and quality of green innovation among polluting firms. First, we construct a partial equilibrium model to describe the asymmetric effects of the CET pilot on green innovation. Then, we conduct empirical examinations based on a sample of polluted firms listed on the A-share market, using patent applications with their citation information. The results show that the CET pilot indeed stimulates the quantity of green innovation through the "leverage effect," which is also moderated by the social responsibility firms assume. However, the green innovation quality and even the overall technological level are significantly inhibited. The results imply that there exists a certain "innovation illusion" in green innovation among polluting firms listed on the A-share market, suggesting that more effective incentive policies should be implemented to ensure both quantity and quality are improved synchronically.

1 | Introduction

Accompanied by a trade-off between environmental protection and economic expansion, green innovation is widely regarded as a critical means of achieving low-carbon development, which reconciles environmental sustainability with long-run competitiveness (Nesta et al. 2014; Sellitto et al. 2020; Zhou et al. 2022). Such a dilemma is particularly pronounced in China, where the traditional growth model has long been characterized by high energy consumption and high emissions (Wang and Lei 2020; Wu et al. 2020; Cui and Cao 2023). Given the scale of environmental challenges, firms face an urgent need to "greenize" their production processes by reducing industrial emissions and upgrading technology.[1] Against this backdrop, the carbon emissions trading (CET) pilot program launched in 2013 has become one of China's most important market-oriented environmental policies (Hu et al. 2020; Chen et al. 2021; Zhu et al. 2022). A central question, however, is whether environmental regulation can induce substantive green innovation.

A substantial body of research suggests that, relative to traditional command-and-control regulation, market-based instruments are generally thought to provide clearer price signals, greater flexibility (Borghesi et al. 2015; Fowlie et al. 2016), and stronger incentives for innovation (Porter and van der Linde 1995; Lanoie et al. 2008; Liu and Sun 2021), rather than merely raising production costs (Ambec and Barla 2002; Mohr 2002). But the innovation effects of CET remain far from settled. On the one hand, the effects may be "mingled." Some studies argue that regulation may crowd out long-term R&D investment by increasing firms' compliance burdens (Palmer et al. 1995; List et al. 2004; Petroni et al. 2019). The empirical evidence likewise points to substantial heterogeneity in the effect of environmental regulation on innovation (Liu and Sun 2021; Han et al. 2023; Brouwers et al. 2016; Borghesi et al. 2015).

On the other hand, most existing studies focus primarily on the "quantity" while paying insufficient attention to the "quality." More green patents do not necessarily imply meaningful technological progress. Under regulatory pressure, firms may just expand patent output through incremental adaptation, strategic patenting, or technological modifications with little "complex knowledge." Such a concern is especially relevant in China. Existing research suggests CET is associated with higher levels of green technological innovation (Han et al. 2023; Sun et al. 2023). But some empirical evidence also implies the possibility of a "bubble" in green patenting in China (Dang and Motohashi 2015; Haddad et al. 2022)—patent applications may be accompanied by a decline in average patent quality (Dang and Motohashi 2015). Since firms respond to environmental regulation by relying on low-value patent applications, the innovation effects of the environmental regulation may be overstated. This trade-off may be particularly acute for heavily polluting firms, which face stricter compliance pressure and must make more difficult choices between short-term performance and long-term green upgrading.

Therefore, this article aims to investigate how the CET pilot affects substantive green innovation among pollution-intensive firms, focusing more on "quality" rather than "quantity." We first develop a partial equilibrium model of firms' decisions over the quantity and quality of green innovation in order to characterize the asymmetric effects of environmental regulation. Empirically, we construct a novel measure of green innovation quality using citation data of green patents in combination with the bipartite network and the nonlinear algorithm. We then implement a difference-in-differences (DID) design using a sample of heavily polluting A-share listed firms to evaluate whether the CET pilot simultaneously promoted both the quantity and the quality of green innovation, or instead generated only an "innovation illusion." Our results show that although the CET pilot significantly increased the quantity of green innovation, it also significantly suppressed the quality. These findings remain robust across a range of tests and provide suggestive evidence of a green innovation "bubble."

This article makes marginal contributions in at least three respects. First, we move beyond the sole focus on innovation quantity and place greater emphasis on the quality of green innovation, thereby providing a more comprehensive assessment of how environmental regulation affects firm innovation. Second, to measure innovation quality, we introduce a novel algorithm based on the bipartite network, which differs substantially from the existing literature that relies on patent citation counts; in doing so, we uncover a "bubble" phenomenon in green innovation. Third, we pay closer attention to a more important category of actors in the Chinese context, namely, heavily polluting firms; by further examining the underlying mechanisms and heterogeneity, we offer more practically relevant policy implications for this group.

The remainder of this article proceeds as follows. Section 2 reviews the related literature. Section 3develops the theoretical framework and hypotheses. Section 4 describes the empirical strategy. Section 5presents the baseline results and robustness checks. Section 6 reports the heterogeneity and mechanism analyses. And Section 7 concludes this article. Figure 1 shows the technology roadmap of this article.

2 | Literature Review

Although environmental regulation raises firms' compliance costs (Jaffe et al. 1995; Palmer et al. 1995), Porter and Linde (1995) argues that well-designed regulation may instead induce innovation by altering the relative payoff structure firms face, thereby encouraging them to improve efficiency, reconfigure production processes, and develop cleaner technologies. A large body of empirical work broadly supports this view, showing that environmental regulation can stimulate innovative activity, even though its effect on overall competitiveness remains contested (Lyu et al. 2020; Han et al. 2023). Formal theoretical contributions likewise suggest that environmental regulation may foster innovation by reshaping firms' incentives to undertake technological adjustment, rather than merely increasing production costs (Ambec and Barla 2002; Mohr 2002).

Such an argument is especially relevant in the case of market-based environmental regulation. Compared with command-and-control ones, market-based instruments typically convey clearer price signals and allow firms greater discretion in choosing how to comply with regulatory targets (Ambec et al. 2013; Lanoie et al. 2011). These instruments, therefore, provide stronger incentives for firms to search for technological solutions that reduce compliance costs while improving environmental performance. In particular, the CET system enables firms to lower compliance costs through emissions abatement and earn additional returns through allowance trading, thereby strengthening the expected payoff to investment in green technologies (Borghesi et al. 2015; Fowlie et al. 2016).

The existing empirical evidence generally supports the incentivizing effect of environmental regulation on green innovation. Early industry-level studies show that environmental innovation responds positively to regulatory pressure to control pollution (Brunnermeier and Cohen 2003). Related work further finds that environmental policies promote advances in renewable energy technologies and pollution-control technologies (Johnstone et al. 2009; Popp 2006). Research on environment-oriented invention similarly indicates that institutional and regulatory pressures can induce firms to engage in green innovation, rather than merely comply passively with environmental requirements (Berrone et al. 2013). More direct evidence comes from the European carbon market, where emissions trading has been shown to increase the number of low-carbon patents among regulated firms (Calel and Dechezleprêtre 2016). Evidence from China likewise suggests that CET programs are associated with higher levels of green technological innovation at both the industry and city levels, although the magnitude of the effect varies across industries and policy settings (Liu and Sun 2021; Han et al. 2023; Sun et al. 2023). More granular firm-level evidence

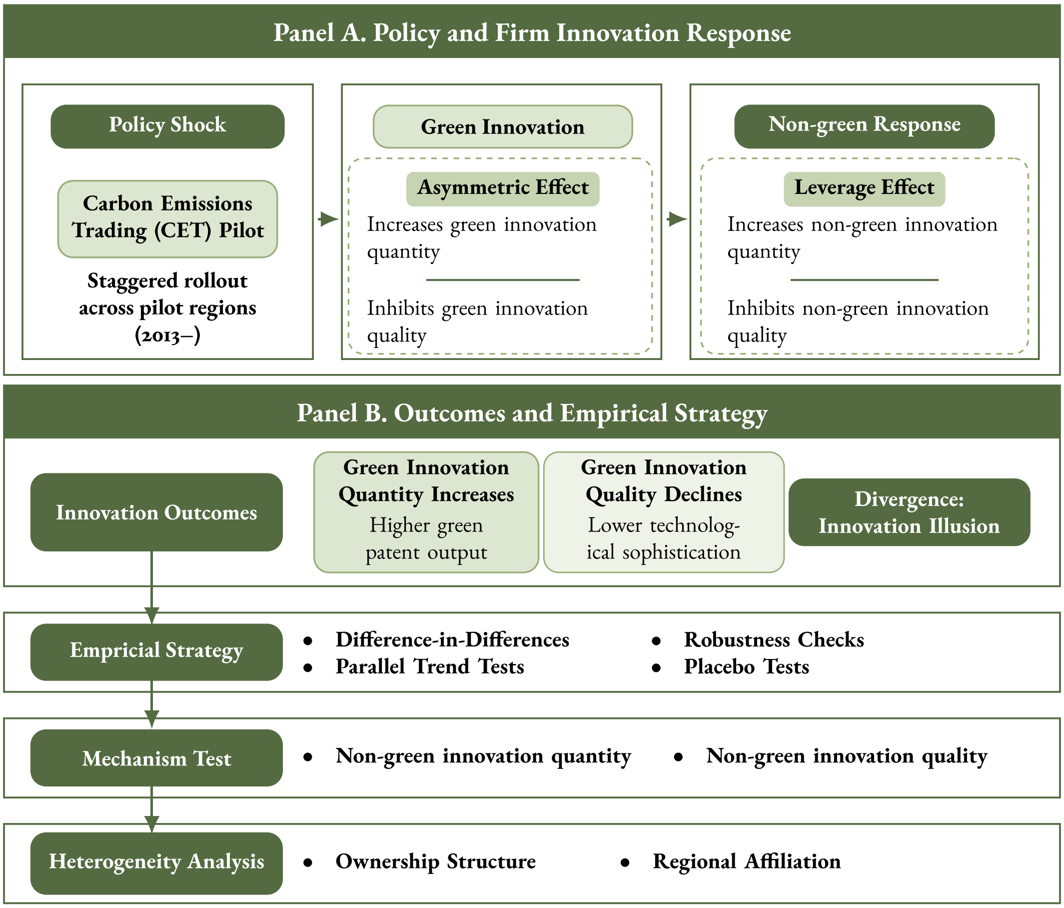


FIGURE 1 | Technology roadmap of this article.

further supports that CET programs stimulate green patenting by listed firms (Liu and Li 2022; Liu and Liu 2023).

Taken together, the literature has established with considerable clarity that environmental regulation, particularly market-based regulation exemplified by the CET, can indeed incentivize firms to undertake green innovation. However, an important limitation of the existing literature lies in the striking neglect of green innovation quality. Most studies rely on patent counts as a proxy for innovation, implying that the effects of environmental regulation on innovation have been examined almost exclusively along the quantity dimension. The problem is that an increase in the number of patents does not necessarily imply a substantive improvement in innovation. Innovation means producing complex knowledge (Balland and Rigby 2017); higher quality innovation is associated with greater technological significance and higher economic value (Trajtenberg 1990; Hall et al. 2000). A larger volume of patent output does not necessarily reflect the production of more complex knowledge, which may instead represent the simple replication of existing knowledge, thereby reducing the technological value embodied in patents. Such a concern is particularly salient in the Chinese context, where policy incentives may increase patenting activity while simultaneously stimulating firms to file relatively low-quality patents (Dang and Motohashi 2015). As a result, the innovation effects of environmental regulations, evaluated solely by patent quantity, may be overstated. Moreover, given the strategic complexity facing heavily polluting firms under environmental regulation, it remains unclear whether such regulation generates green innovation that is both more abundant and higher in quality—or whether it merely induces an expansion of innovation that is more compliance-oriented, incremental, and low in complexity, which has yet to receive a systematic explanation.

3 | Theoretical Framework and Research Hypothesis

3.1 | More and Better Green Innovation?

A substantial body of research shows that the carbon emissions trading (CET) system does stimulate green innovation, particularly in terms of innovation quantity (Liu et al. 2025; Qi et al. 2021; Su 2025). By allowing firms to generate revenue through the sale of surplus allowances, the CET not only raises the opportunity cost of carbon-intensive production but also increases the expected returns to investment in green technologies (Calel and Dechezleprêtre 2016). Internally-driven forces in the cost-benefit structure provide strong incentives for firms to expand green innovation, making it easier to generate large volumes of green technology patents. For pollution costs being internalized into production decisions, firms are compelled to reduce emissions intensity via R&D practice, process upgrading, and the adoption of low-carbon equipment, thereby lowering compliance costs and improving the efficiency of green development (Wang et al. 2021;

Yan et al. 2020; Mardones 2021). Existing empirical evidence likewise confirms that strengthening the additional returns generated by the CET can promote low-carbon technological innovation and green innovation (Qi et al. 2014; Liu and Sun 2021; Hong et al. 2022; Zhang et al. 2021).

Beyond internal incentives, the market-based nature of emissions trading can reinforce firms' incentives for green innovation through information disclosure and external monitoring. Carbon markets help investors identify firms' emissions performance, compliance status, and broader environmental behavior (Gupta and Goldar 2005), meanwhile increasing reputational pressure via media exposure and heightening public risk awareness and monitoring incentives (Capelle-Blancard and Laguna 2010; Tu et al. 2020; Wei et al. 2019; Hao 2023). In seeking support from external stakeholders (Li et al. 2018), polluting firms respond by releasing positive signals through green innovation, under pressure from both environmental costs and increased monitoring intensity. The most direct and observable way of doing so is to increase patent output.

In addition, the CET may increase the quantity of green innovation by easing firms' financing constraints. Innovative activity is typically characterized by high risk, severe information asymmetry, and long investment horizons, and is therefore particularly vulnerable to financial constraints (Hall and Lerner 2010; Cornaggia et al. 2015). By improving the transparency of firms' environmental information, emissions trading lowers the informational barriers faced by outside investors, helps mitigate information asymmetry, and strengthens firms' ability to obtain external financing. Such a mechanism is also supported by existing empirical evidence (Xi and Jia 2025).

Against this background, we propose the first hypothesis:

Hypothesis 1. The CET pilot increase the quantity of green innovation among polluting firms.

Patent output does not necessarily imply an improvement in innovation quality. By introducing new ideas and reshaping the knowledge landscape, technological breakthroughs and disruptive innovations are widely regarded as key drivers of economic growth and industrial transformation (Fortunato et al. 2018; Yang 2025). High-quality innovation should therefore be understood as the production of complex knowledge or, put differently, as technological breakthroughs and disruption. By contrast, the “routine” innovation typically captured by patent counts is conceptually distinct from the “breakthrough” or “disruptive” innovation associated with greater technological significance, originality, or economic value.[2] In this sense, the promoting effect of environmental regulation on innovation documented in much of the existing literature corresponds more closely to the former than to the latter. Empirical evidence likewise suggests that increases in patent filings may, in fact, be associated with lower quality, especially for firms facing strong policy pressure (Yang et al. 2025; Lin et al. 2021).

Environmental regulation may plausibly exert a two-sided effect on innovation quality. Since regulation induces firms to undertake deeper technological upgrading and develop more advanced clean technologies, it may, indeed, improve innovation quality, consistent with the mechanism emphasized by Porter and van der Linde (1995). However, if compliance pressure leads firms to select innovation projects that involve lower uncertainty, fewer resource commitments, and quicker observable outputs, regulation may instead impede innovation quality.

Research on organizational learning and technological search suggests that exploratory and breakthrough innovation is inherently more uncertain than incremental adaptation, and that organizations under pressure tend to exploit familiar search routines and existing technological trajectories rather than engage in more uncertain exploration (March 1991; Fleming 2001). Related work on breakthrough inventions further shows that firms often rely on proximate and established technological paths rather than pursue more radical forms of innovation (Ahuja and Morris Lampert 2001). Under regulatory pressure, especially when firms face mandatory compliance targets or short-term performance constraints, these tendencies may become even more pronounced. Such an interpretation is also consistent with the view that economically valuable innovation often depends on more complex and less ubiquitous knowledge bases, which are more difficult to accumulate and recombine under short-term pressure (Balland and Rigby 2017). Meanwhile, relatively permissive patent examination systems may further encourage strategic behavior by applicants (De Rassenfosse et al. 2021). To obtain generous subsidies and preserve competitive advantage, firms may file large numbers of low-quality patents to crowd out rivals and secure recognition from both the government and the market (Lin et al. 2021).

These considerations imply that environmental regulation may lead to a divergence between innovation quantity and quality. Regulation may stimulate patenting activity by strengthening incentives for compliance-related innovation, while directing innovative effort toward technologies that are less complex and less resource-intensive. This concern is particularly relevant in the context of the CET system and may indicate the presence of a green innovation “bubble.” Existing research on corporate social responsibility likewise suggests that firms may engage in strategic behavior that sacrifices quality to send positive but misleading signals to the market, preserve image and reputation, manage market valuation, and ease financial distress (Geng et al. 2024; Zhang and Jin 2021; Suo and Qiao 2025).

Against this background, we proposes the second hypothesis:

Hypothesis 2. The CET pilot inhibits the quality of green innovation among polluting firms.

3.2 | A Theoretical Model for Green Innovation

To investigate how the CET pilot will increase the quantity of green innovation quantity while inhibit the quantity, here, we construct a partial equilibrium model for describing such an asymmetric effect. All technical details related to model construction are provided in the Appendix.

Following the approach introduced by Mohr (2002), here, we can begin with a simple model. Consider a standard firm profit function Π, with the Pigouvian tax term:

$$\Pi = p\mathscr{Y} - \mathscr{C} - \tau\mathscr{W} \tag{1}$$

where p denotes the product price; $\mathscr{Y}$ denotes the total production; $\mathscr{C}$ denotes the total cost; τ denotes some kinds of the Pigouvian tax; and W denotes the polluting emission level.

The total production $\mathscr{Y}$ can be modeled as the Cobb-Douglas function with a coefficient.

$$\mathscr{Y} = A_p f \tag{2}$$

where A_p denotes the technological level involved in the production process; $f(\cdots)$ denotes production function in the Cobb-Douglas form, incorporating factors including the capital (K), labor (L), intermediate (M) factors, and so forth.

In terms of the technological level A_p, according to Griliches (1998), Bontempi and Mairesse (2008), and Hall et al. (2010), a reasonable parameterization approach is incorporating productivity and patent input factor, which can be expressed as

$$A_p = \gamma_p N_p^{\alpha} \tag{3}$$

where γ_p denote the productivity involved in the production process, which can be captured by production innovation quality; and N_p denote patent input factor involved in the production process, which can be captured by production innovation quantity; α denote the output elasticity.

Assuming a simple case: The firm does not need to consider costs associated with the production process but only needs to consider costs associated with the green innovation process. It implies that the former are sufficiently small relative to the latter; for example, the latter may require substantial fixed costs as well as marginal costs. The costs associated with the green innovation process consist of two components (Sutton 2007; Syverson 2011): (1) the marginal cost related to innovation quantity (the patent input factor N_g) and (2) the fixed cost related to innovation quality (productivity γ_g). Accordingly, the total cost $\mathscr{C}$ can be expressed as

$$\mathscr{C} = cN_g + F \tag{4}$$

where c denotes the unit variable cost associated with the patent input factor and F denotes the fixed cost associated with the productivity, namely, $F(\gamma_g)$.

Worthy noting that we assume the production technology is uniformly convex. Accordingly, whether for innovation output associated with the green R&D process or the production process, the marginal returns to technological improvement are assumed to be diminishing; namely, the output elasticities meet $\alpha, \beta < 0$. In addition, we assume that the marginal cost associated with innovation in the green R&D process is also diminishing, since the R&D process typically involves entry barriers for firms at the outset; thus, $F''(\gamma_g) = \frac{\partial^2 F}{\partial \gamma_g^2} < 0$.

Now, turn to the firm's pollution emission level and the abatement cost it must bear. Consider the following setting. If innovation related to either the production process or the green R&D process is ignored, the firm's inherent level of pollution emissions is $\mathscr{R}$. It means that, in the absence of any production innovation or green innovation, the firm must still pay a certain amount of environmental tax. For simplicity, following Forster (1980) and Selden and Song (1995), we further assume that the firm's two processes affect pollution emissions and abatement costs in the following way:

i. Innovation associated with the production process always generates environmental pollution; and
ii. Innovation associated with the green R&D process always reduces pollution.

Accordingly, this setting can be modeled as follows:

$$\mathscr{W} = \mathscr{R} + P(A_p, g) - G(A_g, h) \tag{5}$$

where $P(\cdots)$ denotes the pollution level determined by technological level A_p and production function $g(\cdots)$; $G(\cdots)$ denotes the governance level determined by technological level A_g and production function $h(\cdots)$; $g(\cdots)$ and $h(\cdots)$ denote the production function involved in pollution emission process and pollution governance process, respectively, also incorporating input factors including capital, labor, and intermediate inputs, etc.

It is important to note that we consider only two processes here, namely, the production process and the green R&D process. The production process is associated with the firm's representative product. Producing products necessarily pollutes the environment. A higher technological level in such a process just represents an improvement in production efficiency, rather than a reduction in pollution; even, improved production efficiency exacerbates pollution emissions. By contrast, the green R&D process is unrelated to the firm's representative product and does not generate profit. Its sole role is to reduce pollution emissions and thereby lower the firm's additional abatement costs. Moreover, the firm must also incur additional variable and fixed costs for such a process. Therefore, the firm faces a trade-off in its decisions between the production process and the green R&D process.

We can investigate the following specific forms of the pollution level:

$$P(A_p, g) = \left(\phi A_p g\right)^{\eta} \tag{6}$$

and the governance level:

$$G(A_g, h) = \left(\varphi A_g h\right)^{\theta} \tag{7}$$

where ϕ denotes the "productivity" involved in the pollution emission process and φ denotes the counterpart in pollution governance process. Similar to the technological level of production process, corresponding counterpart in green R&D process can also be parameterized as

$$A_g = \gamma_g N_g^{\beta} \tag{8}$$

where β also denotes the output elasticity. In addition, within both the pollution emission and governance processes, we allow the non-constant returns to scale, which is realized by setting the scale parameter η and θ (Hallak and Sivadas 2009, 2013). Here, we continue to assume here that the output associated with pollution emissions and the pollution abatement process should also exhibit convexity, namely, diminishing marginal returns. Thus, both scale parameter η and θ meet $0 < \eta, \theta < 1$. It is worth mentioning that we assume both the fixed-cost function associated with the green R&D process and the corresponding output function of the pollution abatement process are convex. Here, we further assume that the former is more strongly convex than the latter.[3]

Put together, the profit function of the firm can be expanded as

$$\begin{aligned}\Pi(\gamma_p,\gamma_g,N_p,N_g) &= p\gamma_p N_p^{\alpha} f - cN_g - F(\gamma_g) \\ &\quad - \tau\left[\mathscr{R} + \left(\phi\gamma_p N_p^{\alpha} g\right)^{\eta} - \left(\phi\gamma_g N_g^{\beta} h\right)^{\theta}\right]\end{aligned} \tag{9}$$

For investigating the optimal patent factor input involved in green R&D process, we can calculate the derivative of Π with respect to N_g and let it equal to 0:

$$\frac{\partial \Pi}{\partial N_g} = -c + \tau\gamma_g^{\theta}\beta\theta N_g^{\beta\theta-1} = 0 \tag{10}$$

Then we can obtain the optimal N_g, and we show that the derivative of N_g with respect to τ will be positive, namely:

$$\frac{\partial N_g}{\partial \tau} > 0 \tag{11}$$

which means that, as environmental regulation becomes more stringent, the firm's optimal decision (holding other factors constant) is to choose a greater level of green innovation (patent) input in terms of quantity. Thus, we propose the following proposition:

Proposition 1. *As the CET pilot implements, the quantity of green innovation will increase.*

Now, turn to investigate the optimal decision on green innovation quality the firm should choose. For this purpose, we calculate the derivative of Π with respect to γ_g, and let it equal to 0:

$$\frac{\partial \Pi}{\partial \gamma_g} = -F' + \tau\theta\gamma_g^{\theta-1}N_g^{\theta\beta} = 0 \tag{12}$$

Then we can also obtain the optimal γ_g, and we show that the partial derivative of γ_g with respect to τ will be negative, namely:

$$\frac{\partial \gamma_g}{\partial \tau} < 0 \tag{13}$$

which implies that, as environmental regulation becomes more constringent, the firm will choose a lower level of innovation quality as its optimal decision. Thus, we propose the following proposition:

Proposition 2. *As the CET pilot implements, the quality of green innovation will decline.*

4 | Research Design

4.1 | Core Explanatory Variable: Innovation Quantity and Quality

The explained variable used in this article refers to the level of green innovation, including quantity and quality dimensions. Thus, first, to measure the quantity aspect of green innovation among polluted firms, this article acquires data on patent applications for inventions and utility models for firms listed on the A-share market from the China National Intellectual Property Administration (CNIPA). Then, based on the WIPO International Patent Classification (IPC) Green Inventory, this article conducts matching for green patents and aggregates and logarithmizes the total number of patents that meet the criteria, resulting in 2 indicators: green invention applications (GNinv) and green utility applications (GNuti). Finally, the logarithm of the total number of green inventions and utility model patents is used to obtain the green innovation applications (GNinn) indicator. These three indicators will be used to measure the quantity dimension of firms' green innovation.

To measure the "quality" of green innovation more accurately and reasonably, this article considers adopting complex network analysis tools, drawing on the approaches of Balland and Rigby (2017), Zhang and Zhang (2024), and Zhang (2024). Specifically, this article first constructs the firm-patent bipartite network based on the citation information of green patents, and then constructs the indicator of Generalized Knowledge Complexity (GKC) by combining the Fitness and Complexity (FC) algorithm with the Matrix-estimation exercise to measure the "quality" of green patents.

First, consider constructing the firm-patent bipartite network of a certain year by mapping the citation information of green patents, whose adjacency matrix corresponds to

$$\boldsymbol{M} = \begin{pmatrix} M_{11} & M_{12} & \cdots & M_{1t} \\ M_{21} & M_{22} & \cdots & M_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ M_{f1} & M_{f2} & \cdots & M_{ft} \end{pmatrix} \tag{14}$$

where M_{ft} denotes the patents' citation situation (referring to the International Patent Classification) within the technology domain t of firm f, which can usually be calculated by the Revealed Comparative Advantage (RCA) introduced by Balassa (1965).

Within the bipartite network, issues of the knowledge complexity can be described by determining two properties—X_f and Y_t, which represent the complexity properties of firms and technology domains respectively. The property X_f can be expressed as a function composed of property $Y_1, Y_2, ..., Y_t$ and the adjacency matrix M_{ft} of the bipartite network, and vice versa. Then, studying economic problems is just equal to solving the coupled equations below:

$$\begin{cases} X_f = f(Y_1, Y_2, \cdots, Y_t, M_{ft}) \\ Y_t = g(X_1, X_2, \cdots, X_f, M_{ft}) \end{cases} \tag{15}$$

In terms of bipartite network analysis, X_f denotes the "complicacy" of a firm node, which, in the context of this article, represents the innovation quality that firm f creates through R&D activities. Y_t refers to the "complicacy" of a technology domain node, representing the innovation quality that technology domain t contains contributed by firms. Equation (15) implies that: innovation quality of firms X_f depends on the innovation quality of its connected technology domain nodes ($Y_1, Y_2, \ldots, Y_t$) and their linkages (M_{cp}); similarly, innovation quality of technology domains Y_p depends on the innovation quality of its connected firm nodes ($X_1, X_2, \ldots, X_c$) and their linkages (M_{cp}).

By solving Equation (15) with a specific algorithm, $\hat{X}_f$ and $\hat{Y}_t$ can be obtained, which are referred to as "complexities," distinguishing them from the aforementioned "complicacy"—the latter denotes the attribute of nodes, while the former is the quantification of the latter.

In fact, in this article, we solve for X_f and Y_t using the Fitness and Complexity (FC) algorithm proposed by Tacchella et al. (2012), which sequentially computes the firm proximity matrix $N_{ff'}$:

$$N_{ff'} = \frac{\sum_{f'} M_{ft} M_{f't}}{(s_{t'})^2 s_{f'} s_f} \tag{16}$$

On the basis of Equation (16), the indicator of GKC can be constructed by

$$GKC_{ft} = \left(\sum_{i=1}^{2} \lambda_i^{\boldsymbol{N}_t} \left(v_{f,i}^{\boldsymbol{N}_t} \right)^2 \right)^2 + 2 \sum_{i=1}^{2} \left(\lambda_i^{\boldsymbol{N}_t} \right)^2 \left(v_{f,i}^{\boldsymbol{N}_t} \right)^2 \tag{17}$$

where $\lambda_1^{\boldsymbol{N}_t}$ and $\lambda_2^{\boldsymbol{N}_t}$ are the largest two eigenvalues of the proximity matrix $\boldsymbol{N}_t$ and $v_{c,1}^{\boldsymbol{N}_t}$ and $v_{c,2}^{\boldsymbol{N}_t}$ are the eigenvectors corresponding to $\lambda_1^{\boldsymbol{N}_t}$ and $\lambda_2^{\boldsymbol{N}_t}$, respectively.

Actually, GKC can be seen as a derivative indicator of Eigenvector Centrality, with a topological meaning similar to the latter; this article will treat GKC as the "quantification" of innovation quality in the following text to conduct further empirical studies. The higher the GKC is, the higher the quality of innovation will be.

As mentioned above, the firm-patent bipartite network is constructed based on patents' citation information. Therefore, we construct bipartite networks based on the citation information of green patent grants and applications of different firms for each year. Then, we calculate the GNcom indicator, namely the complexity of green innovation applications, to measure green innovation quality. Worthy of mention, GNcom is calculated using GKC quantiles (i.e., ranking percentages of GKC) to eliminate self-correlation and heteroscedasticity.

4.2 | Explained Variable: The CET Pilot

The core explanatory variable in this article concerns the treatment of the CET pilot across heterogeneous environmental regulatory intensities. Regarding the implementation of the CET pilot, it commenced in 2011 and unfolded in two stages: The first stage (2011–2020) involved the pilot in eight provinces and municipalities—Beijing, Shanghai, Tianjin, Chongqing, Hubei, Guangdong, Shenzhen, and Fujian; the second stage began in 2021 with the establishment of a nationwide carbon emission trading market. Accordingly, this article defines a dummy variable, the CET pilot, which indicates whether a specific firm participated in the CET market in a given year.

In addition, the core explanatory variable captures the intensity of environmental regulation. To quantify differences in implementation effort or environmental attention given to environmental regulations across regions, this article considers incorporating the intensity of environmental regulations (represented by "regulation"). Following the approach of Qin et al. (2021), this article employs text analysis by way of web scraping techniques in Python. And then, the frequency of occurrence of keywords related to "environmental protection" in the government work reports of various prefecture-level cities is calculated as a proportion of the total word count in the corresponding reports, serving as a measure of the intensity of environmental regulations.

The core explanatory variable is finally constructed as the interaction between the CET pilot dummy variable and the intensity of environmental regulations.

4.3 | Other Variable Specification

Additionally, this article considers the introduction of control variables that may affect green innovation. Drawing on previous empirical research on green innovation, this article incorporates the following eight control variables in the econometric model: firm asset (asset), operating revenue (revenue), firm age (age), flow ratio (flowratio), turnover capability (turnover), dividend per share (DPS), equity incentive (incentive), leverage ratio (lever), EPS growth (EPS_g), and return on equity (ROE). The abbreviations, names, and descriptions of the variables are shown in the Appendix.

To ensure an adequate sample for the CET pilot, this article selects the period from 2007 to 2020. Additionally, referring to the Industry Classification Directory for Environmental Inspection of Listed Companies (Index Number: 000014672/2008-00519), a total of 1269 firms from polluting industries listed on the A-share market are selected as the sample. The descriptive statistics of variables are presented in the Appendix, and worth mentioning, the sample data have been winsorized at a 1% level in both lower and upper tail.[4]

4.4 | Econometric Model

To investigate the relationship between heterogeneous environmental regulations and green innovation of polluting firms, this article still places the research within the framework of a quasi-natural experiment. Specifically, the article considers constructing a Difference-in-Difference-like model and then conducts regressions with such a generalized staggered DID expression:

$$\text{GreenInnov}_{it} = \alpha + \beta \text{EnvirReg}_{it} + \sum_{k=1}^{K} \gamma_k X_{k,it} + \lambda_t + \mu_i + \varepsilon_{it} \tag{18}$$

where GreenInnov_{it} denotes green innovation performance of polluting firms, in "quantity" level (corresponding to GNinn_{it}, GNinv_{it}, and GNuti_{it}) or "quality" level (corresponding to GNcom$_{it}$); EnvirReg$_{it}$ denotes heterogeneous environmental regulation in terms of CET pilot; $X_{k,it}$ denotes control variables that may affect green innovation; λ_t and μ_i denote time and individual fixed effects.

5 | Result

5.1 | Innovation Quantity

This article first investigates whether the implementation of heterogeneous CET pilots enhances polluting firms' performance in terms of the "quantity" of green innovation. Before conducting formal regressions, the article conducts parallel tests on the treatment variable for each type of explained variable. The results depicted in Figure 2 indicate that there exists no significant difference within the pre-treatment period. Ensuring that both control and treatment groups pass the parallel trend tests, this article applies a DID regression using three distinct indicators (GNinn, GNinv, and GNuti) as explained variables, and the results are shown in Table 1. Furthermore, regression results with all control variables included and excluded are presented in the table.

The results indicate that the coefficients of *treat* × *post* within each group are all significantly positive at the 5% level, indicating that the implementation of China's CET pilot can promote green innovation among polluting firms listed on the A-share market at a sufficiently high level of significance. Moreover, there are no significant differences in the regression results across different explained variables or within the same variable, suggesting the robustness of the baseline regression findings. Specifically, after excluding all control variables, the coefficients of the core explanatory variable are 23.395, 12.492, and 20.334 for each group. This implies that for every 0.01-unit increase in environmental regulation (namely, for every 1 percentage point increase in the frequency of environmental regulation related keywords in the government reports of the regions where CET pilot firms are located), the total number of green patents are expected to increase by about 23.4%, the number of green invention patents by about 12.9%, and the number of green utility model patents by about 20.3%. Moreover, after including the full set of control variables, the coefficients of the core explanatory variable remain largely unchanged across all groups. This suggests that we have largely identified the "clean" innovation-inducing effect of environmental regulation.

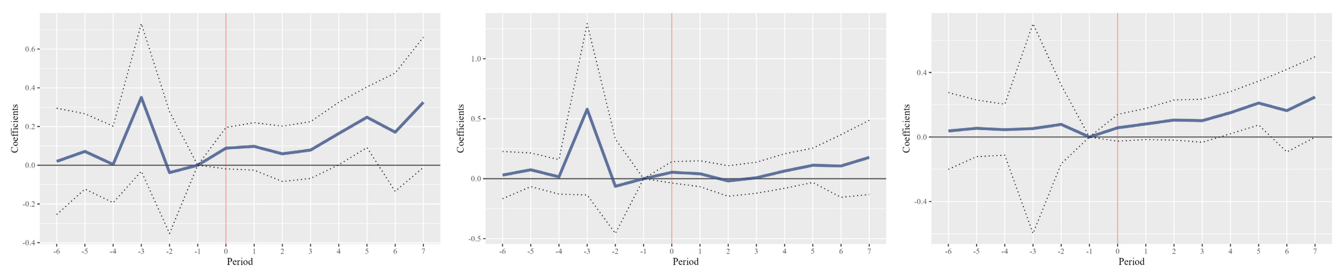
5.2 | Robustness Check

Subsequently, this article conducts robustness checks to ensure the reliability of the baseline regression results. Specifically, seven robustness checks were conducted across three explained variables. Each robustness test category assesses two scenarios, both with and without control variables, as shown in Figure 3. The robustness checks include:

- Core explanatory variable replacement ("Core X Replacement"): The core explanatory variable used in baseline regressions interacts with CET pilot implementation and the frequency of environmental-related keywords in government reports. Here, this article substitutes the core explanatory variable with "CET pilot," indicating whether a firm implemented the CET pilot in a given year.
- Time-lag and anticipative effects ("Lead/Lag Effect"): The core explanatory variable is replaced by its lagged 1 term and led 1 term to investigate time-lag and anticipatory effects, respectively.
- Explained variable replacement ("Y Replacement"): The baseline regression employs the number-based indicators of patents and utility models of green invention filed by firms annually. Here, this article substitutes these three dependent variables with the corresponding number-based indicators of granted patents (GNinn_g, GNinv_g, GNuti_g). It is worth mentioning that lagged two terms are employed to account for the delay between the patent application and grant.
- Additional interactive fixed effects ("Additional FE"): Additional interaction fixed effects, including time-province and time-city interactions, are introduced into the regressions, respectively.
- Propensity Score Matching ("PSM"): Samples are matched based on Caliper-Nearest matching method and Generalized Linear Model (GLM) distance prior to conducting DID regression.

Finally, this article conducts placebo tests by employing 500 times bootstrap re-sampling regarding the core explanatory variable, with results shown in Figure 4. Notably, only results incorporating all control variables are presented here.

The robustness test results in Figure 3 generally support the baseline regression conclusions, confirming that CET



(a) (b) (c)

FIGURE 2 | Results of parallel tests.

TABLE 1 | Results of baseline regressions.

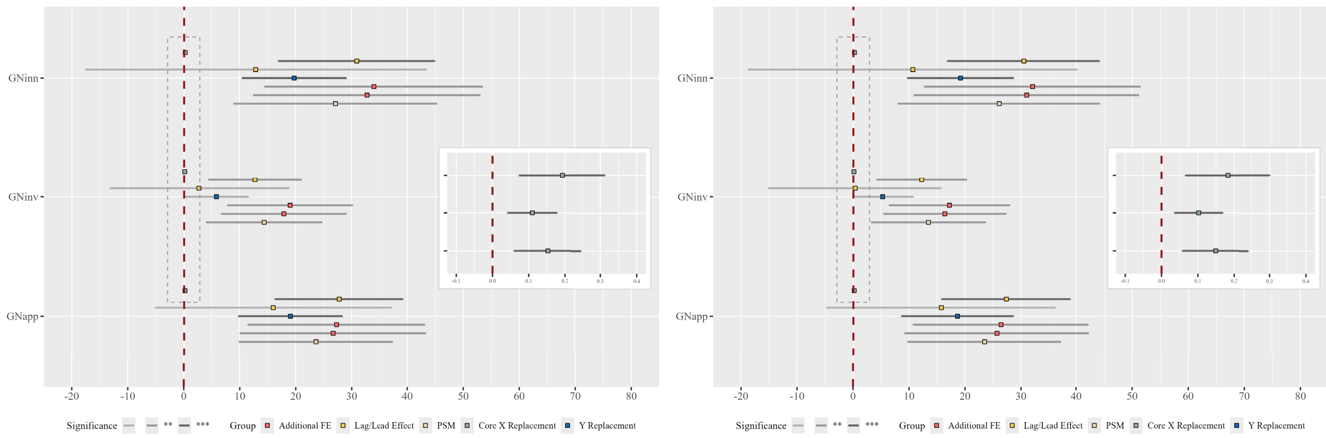
	Green innovation quantity					
	GNinn		GNinv		GNuti	
Variable	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
regulation	23.397**	22.097**	12.492**	11.313**	20.334**	19.838**
	(9.445)	(8.997)	(5.519)	(5.188)	(7.144)	(6.798)
asset		−0.001		−0.027		0.023
		(0.033)		(0.027)		(0.027)
revenue		0.070		0.070*		0.024
		(0.042)		(0.036)		(0.032)
age		0.387*		0.303***		0.314*
		(0.186)		(0.085)		(0.174)
flowratio		0.015**		0.014**		0.003
		(0.006)		(0.006)		(0.006)
turnover		−0.053		−0.041		−0.027
		(0.042)		(0.026)		(0.037)
DPS		0.114*		0.068*		0.021
		(0.063)		(0.038)		(0.056)
incentive		−0.023		−0.030		−0.021
		(0.031)		(0.020)		(0.026)
lever		−0.147		−0.042		−0.176**
		(0.118)		(0.098)		(0.072)
EPS_g		−0.006		−0.001		−0.008
		(0.008)		(0.006)		(0.006)
ROE		−0.111		−0.006		−0.046
		(0.263)		(0.236)		(0.152)
Observations	846	846	846	846	846	846
Adj. R2	0.581	0.584	0.583	0.585	0.496	0.498
Time FEs	✓	✓	✓	✓	✓	✓
Individual FEs	✓	✓	✓	✓	✓	✓

*$p < 0.1$.
**$p < 0.05$.
***$p < 0.01$.

pilot implementation significantly improves polluting firms' green innovation performance. Additional findings include the following: First, the implementation of the CET pilot significantly enhances the quantity of green innovation, even without accounting for heterogeneous environmental regulatory intensity. Second, the coefficients for the anticipatory effect (led by one term) are statistically insignificant at the 10% level across all six groups, indicating no significant anticipatory effects. In contrast, the significance of the lagged one term just confirms a delayed impact. Furthermore, regression results with granted green innovation patents demonstrate that environmental regulation "substantively" enhances firm innovation outcomes, reflected in "effective" output. Lastly, the results of placebo tests reveal a substantial deviation of the coefficient in the baseline regression from its sampling distribution based on bootstrap resampling, validating the baseline estimates of policy effects.

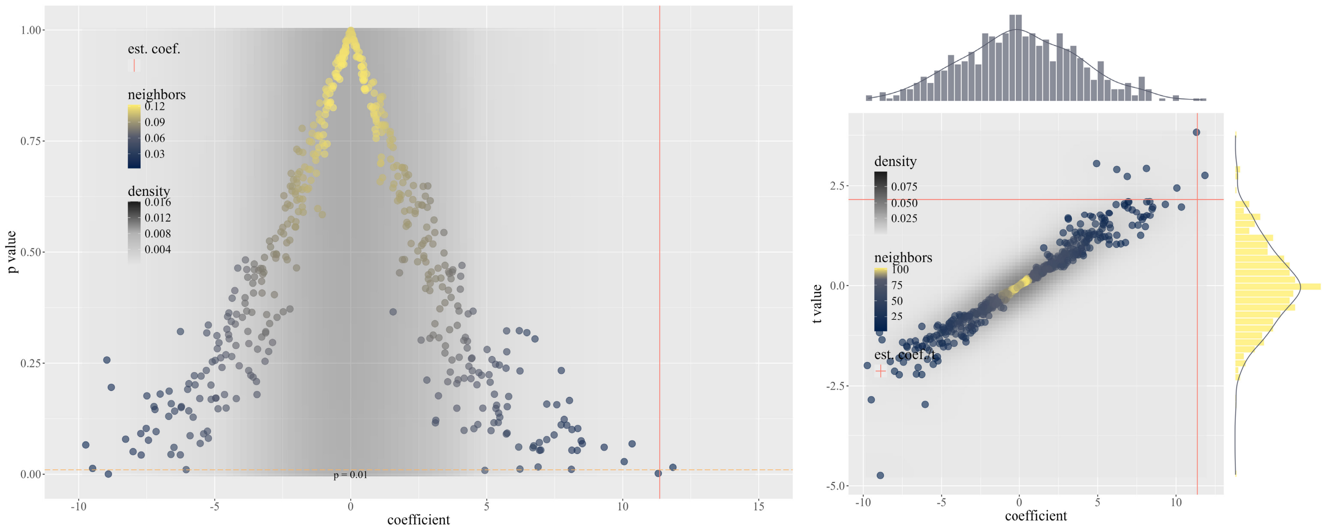
5.3 | Innovation Quality

Although the stimulation of the carbon emission trading pilot on the "quantity" of green innovation in polluting firms has been verified above, this article aims to further evaluate the "quality" of green innovation, thereby examining whether the CET pilot continues to promote its "quality."



(a) (b)

FIGURE 3 | Results of robustness checks.



(a) (b)

FIGURE 4 | Results of placebo tests.

As mentioned above, in the face of environmental regulations, firms, especially polluting ones, do have the motivation to reinforce their R&D practices to improve their pollution governance capabilities. However, to quickly meet environmental regulatory requirements, reduce the operating costs of environmental governance, and mitigate the sunk costs of R&D, thereby reducing potential loss risks, imitating existing technologies is usually a preferred choice for a firm. While a large number of firms seek to improve their environmental governance through such an approach, there will be a significant outputting of patents with "homogeneity" in the field of environmental governance. However, breakthrough innovations are rarely seen. Actually, such significantly "produced" patents with homogeneity do not require a high level of advanced factors, such as complex knowledge, capital investment, and so forth., which means their "quality" is relatively low. On the other hand, breakthrough innovations imply the ability to deliver sufficient economic value to firms, indicating a higher "quality" of such patents.

To assess the impact of heterogeneous CET pilot implementation on the green innovation "quality" of polluting firms, this article redefines the explained variables and re-conducts DID regressions. Similar parallel tests precede formal regression, with results in Figure 5 again indicating no significant pre-treatment difference exists. After ensuring parallel trends, this article uses GNcom as the explanatory variable for green innovation quality, with results shown in Table 2. Columns (1) and (2) present results of regressions using the dummy variable "CET pilot" as the core explanatory variable, while Columns (3) and (4) present results using the primary core explanatory variable "regulation" used in the previous text. Regression results with and without all control variables are presented.

The regression results indicate significantly negative coefficients across the four groups at least at the 10% level. Given that GNcom is derived from GKC quantiles, higher GNcom values indicate higher patent quality. Therefore, these findings suggest that CET pilot implementation significantly inhibits the

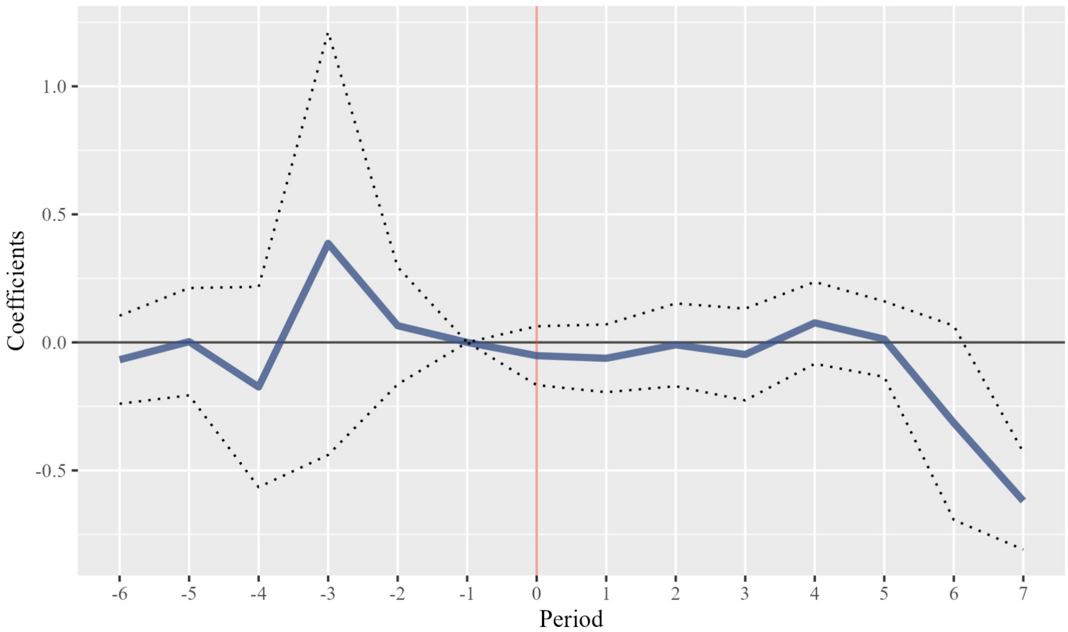


FIGURE 5 | Result of parallel test based on GNcom.

TABLE 2 | Results of regressions based on green innovation complexity.

	Green innovation quality			
Variable	**(1)**	**(2)**	**(3)**	**(4)**
CET pilot	−0.121**	−0.140**		
	(0.053)	(0.059)		
regulation			−14.542*	−16.676*
			(8.168)	(8.185)
Control variables	✗	✓	✗	✓
Observations	846	846	846	846
Adj. R_2	0.576	0.573	0.573	0.570
Time FEs	✓	✓	✓	✓
Individual FEs	✓	✓	✓	✓

*$p < 0.1$.
**$p < 0.05$.
***$p < 0.01$.

"quality" of green innovation among polluting firms, such an effect that persists even when accounting for heterogeneous environmental regulations. Specifically, after excluding all control variables, the coefficient of CET pilot is −0.121, while the coefficient of regulation is −14.542. This implies that, when a firm is included in the CET pilot, the percentile rank of its green innovation quality (GKC quantile) is expected to decline by 12.1 percentage points (which indicates an improvement in ranking). Likewise, when the frequency of keywords related to environmental regulation in the local government work report increases by 1 percentage point, the percentile rank of the firm's green innovation quality is expected to decline by about 14.5 percentage points. After including the full set of control variables, the regression results remain largely unchanged. This also suggests that we have, to a large extent, identified the clean policy effect of environmental regulation on the quality of green innovation.

6 | Mechanism Test and Heterogeneity Analysis

6.1 | Leverage Effect

Intuitively, an important issue worth further attention is whether the observed impact of the CET pilot on green innovation among polluting firms, whether it manifests as a positive or negative effect on innovation quantity or quality, is driven by a "leverage effect" or a "crowding-out effect." In other words, does the CET pilot influence other dimensions of firms' innovation activities? On the one hand, if the improvement in green innovation performance stems from an overall increase in R&D investment across all departments—where greater input into non-green R&D indirectly stimulates green innovation—then the CET pilot manifests as a "leverage effect" on green innovation. On the other hand, if firms enhance their green innovation performance at the expense of other R&D activities, reallocating resources away from non-green innovation, the CET pilot's impact would reflect a "crowding-out effect."

Returning to the model constructed earlier, we can further infer that the impact of the CET pilot on green innovation should take the form of a "leverage effect." That is, the innovation effects on both green and non-green activities, whether manifested as an increase in quantity or a decline in quality, should move in the same direction. Specifically, we first investigate the optimal decision of firms on non-green innovation, namely, we calculate the derivative of Π with respect to N_p and γ_p and let it be 0, respectively:

$$\begin{cases} \partial\Pi/\partial N_p = p\gamma_p \alpha N_p^{\alpha-1} - \tau\gamma_p^{\eta}\alpha\eta N_p^{\alpha\eta-1} = 0 \\ \partial\Pi/\partial\gamma_p = pN_p^{\alpha} - \tau\eta\gamma_p^{\eta-1}N_p^{\alpha\eta} = 0 \end{cases} \tag{19}$$

Here, we will obtain the optimal N_p and γ_p. Then, we calculate the partial derivatives of N_p and γ_p with respect to N_g, and we show that they will both be positive:

$$\begin{cases} \partial N_p/\partial N_g > 0 \\ \partial\gamma_p/\partial\gamma_g > 0 \end{cases} \tag{20}$$

which means that, as green innovation patent increases, non-green one also rises accordingly, and as green innovation quality declines, non-green one also declines accordingly. Hence, the green innovation exhibits a "leverage effect" on non-green one, both at the quantity and quality levels.

Empirically, this article employs three indicators capturing firms' non-green or non-low-carbon innovation levels as dependent variables: non-green innovation (nGNinn), non-green invention (nGNinv), and non-green utility model (nGNuti). These indicators correspond directly to the previously used green indicators, calculated as the logarithm of firms' non-green or non-low-carbon patent applications in a given year. Regarding the quality of green innovation, this article uses an indicator (Tcom) that reflects the overall innovation quality of firms (without distinguishing between green and non-green patents), measured using the previously described construction method of the GKC indicator based on patent citation data within a given year. Regression results are presented in Table 3, which provides estimates both with and without control variables. Additionally, due to table structure considerations, the table displays only the regression results for the quantity dimension, with non-green innovation (nGNinn) as the dependent variable.

The regression results indicate that, in terms of quantity, the coefficients of regulation are significantly positive at the 5% level. This finding implies that implementing the CET pilot has indeed promoted non-green, non-low-carbon innovation among polluting firms in the pilot regions. In other words, the CET pilot primarily enhances green innovation via a "leverage effect." Conversely, in the quality dimension, the coefficients of regulation are significantly negative at the 10% level, suggesting that the CET pilot not only negatively impacts the quality of green innovation but also reduces the overall innovation quality of polluting firms. It indirectly confirms the existence of a "leverage effect" in firms' green innovation behaviors due to the CET pilot policy. The minor differences between the coefficients of regulation with and without control variables also further attest to the robustness of these findings.

TABLE 3 | Results of mechanism test.

	Green innovation			
	Non-green innovation quantity		Overall innovation quality	
Variable	(1)	(2)	(3)	(4)
regulation	30.290**	26.549**	−11.822*	−12.531*
	(11.745)	(11.724)	(6.874)	(6.789)
Control variables	✗	✓	✗	✓
Observations	594	594	594	594
Adj. R^2	0.659	0.661	0.003	0.037
Time FEs	✓	✓	✓	✓
Individual FEs	✓	✓	✓	✓

*$p < 0.1$.
**$p < 0.05$.
***$p < 0.01$.

6.2 | Heterogeneity Analysis

As mentioned earlier, the extent of social responsibility that firms should bear will affect the role of environmental regulations in promoting green innovation. As state-owned enterprises (SOEs), especially in China, it is undeniable that they ought to fulfill considerable social responsibilities, for the fulfillment of social responsibility by SOEs is a fundamental requirement for their establishment with state funding and is also their highest mission. Therefore, compared to non-state-owned enterprises (non-SOEs), SOEs are more likely to already have a higher level of green innovation even before the CET pilot. It implies that non-state-owned enterprises may face more significant environmental governance performance pressures than state-owned enterprises, thereby incentivizing at least quantitative improvements in the firm's green innovation. However, as previously discussed, performance pressures arising from environmental governance typically compel firms to quickly meet environmental regulatory requirements, forcing them to produce relatively homogeneous R&D outputs rather than breakthrough innovations. Consequently, the quality of green innovation among non-SOEs may actually decline.

The population size and economic scale of the Yangtze River Economic Belt account for a significant portion of China's

TABLE 4 | Results of grouping regressions based on ownership attribute.

	Green innovation							
	Non-state-owned		State-owned		Non-Yangtze Belt		Yangtze Belt	
Variable	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
regulation	34.338**	−28.397**	7.918	−13.350	32.966***	−18.322**	−6.029	−5.599
	(13.440)	(10.958)	(7.816)	(10.025)	(9.159)	(6.826)	(11.863)	(26.695)
Control variables	✓	✓	✓	✓	✓	✓	✓	✓
Observations	350	496	350	255	350	496	350	255
Adj. R^2	0.458	0.643	0.669	0.537	0.629	0.464	0.558	0.540
Time FEs	✓	✓	✓	✓	✓	✓	✓	✓
Individual FEs	✓	✓	✓	✓	✓	✓	✓	✓

*$p < 0.1$.
**$p < 0.05$.
***$p < 0.01$.

resources. Given its prominent ecological position, companies belonging to the Yangtze River Economic Belt are expected to embrace sustainable development, which means that these companies, regardless of their participation in the CET pilot, should fulfill a series of social responsibilities, including environmental governance to a certain extent, and thereby, inherently possess considerable green innovation. Similarly, it suggests that firms outside the Yangtze River Economic Belt may face stronger quantitative incentives to pursue green innovation. Yet, as previously indicated, environmental regulatory pressure aimed at improving green innovation performance may negatively impact the quality of R&D innovation. Hence, firms outside the Yangtze River Economic Belt might also encounter declines in the quality of their green innovations.

Therefore, this article believes that the ownership attribute and regional affiliation of polluting firms may significantly influence the effect of the carbon emissions trading pilot on their green innovation, regardless of whether it is at the "quantity" level or "quality" level. Thus, the article conducts analyses based on grouping regressions. To be more specific, the original sample is divided into non-state-owned and state-owned samples based on ownership attribute, and into non-Yangtze River Economic Belt and Yangtze River Economic Belt samples based on regional affiliation. Then, DID regressions will be conducted respectively for each group, and the results based on indicators of green innovation quantity (GNinn) and quality(GNcom) are shown in Table 4.

The results in Table 4 align with the expectations of this article. First, regarding the quantity of green innovation, the coefficient of regulation in non-SOE samples is significantly positive at a sufficiently high level of significance; whereas, the coefficient in SOE samples is insignificant at the 10% level. It demonstrates that the CET pilot's positive impact on the quantity of green innovation differs significantly across these two sample groups, highlighting stronger incentives for non-state-owned polluting firms. Meanwhile, the coefficient of regulation for firms outside the Yangtze River Economic Belt is significantly positive at a high significance level; whereas, it remains insignificant at the 10% level for firms within the Yangtze River Economic Belt. It indicates that the CET pilot provides stronger quantitative incentives for green innovation among polluting firms located outside the Yangtze River Economic Belt. In fact, these results implicitly underscore the crucial moderating role of corporate social responsibility in enhancing this incentivizing effect. The CET pilot indeed promotes innovation among non-state-owned firms and those outside the Yangtze River Economic Belt. However, state-owned firms and those within them typically bear greater social responsibilities and thus encounter less additional environmental governance pressure from the CET pilot, thereby weakening their incentives to enhance green innovation performance.

Secondly, results regarding the quality of green innovation also support our expectations. The coefficient of regulation is significantly negative at sufficiently high levels for both non-SOE samples and firms outside the Yangtze River Economic Belt. Conversely, the coefficients remain insignificant at the 10% level for SOEs and firms within the Yangtze River Economic Belt. Indeed, these regression results on green innovation quality reinforce the pivotal role of corporate social responsibility. Green innovation behaviors are influenced by the level of social responsibility enterprises assume. Due to their inherent social responsibility obligations, SOEs and Yangtze River Economic Belt firms may not face urgent pressure to meet compliance standards quickly, implying that these firms experience reduced pressures on green innovation performance. Consequently, they will be able to avoid diverting corporate resources toward homogeneous patent outputs, thereby preventing a decline in innovation quality.

7 | Concluding Remarks

In the context of the low-carbon economy, it cannot be denied that green innovation is vital. While many firms have gradually dedicated themselves to the R&D of green- and low-carbon technologies, there have indeed been tangible outcomes in terms of patent output. However, it is more crucial for firms or nations to avoid "patent bubbles" and "innovation illusions" during the phase of the explosive growth of patent output.

To understand how environmental regulations influence firms' green innovation, this article conducts empirical studies on examining the impact of the carbon emissions trading pilot on the "quantity" and "quality" aspects of green innovation in polluting firms listed in the A-share, by employing DID regressions within the quasi-natural experiment framework, based on patent applications data of firms with citation information. The results indicate that the carbon emissions trading pilot can indeed stimulate green innovation of polluting firms in terms of "quantity" through the "leverage effect." However, at the "quality" level, the carbon emissions trading pilot merely hinders green innovation among polluting firms; moreover, it significantly impedes their overall technological advancement. In fact, such an inhabitation suggests that polluting firms may commonly exhibit an "innovation illusion" in response to environmental regulatory pressures; moreover, research findings also suggest that corporate social responsibility plays a moderating role in alleviating environmental compliance pressure, thereby indirectly influencing firms' innovation behavior.

The policy implications of this article are straightforward but of great concern. In the context of growing attention to environmental issues, developing countries with underdeveloped environmental governance must establish effective government-funded incentive policies that ensure simultaneous, sustainable improvements in both the quantity and quality of green innovation. The focus of the governmental policies should shift from simply encouraging quantity growth to incentivizing the creation of high-quality patents; at the same time, region-specific governmental policies should be implemented based on the differences in regional innovation capabilities; furthermore, attention should be paid to coordinating policies, which refers to strengthening intellectual property protection systems, establishing a fair market competition environment, and adjusting the openness strategy. Not only green innovation under environmental regulations but also policies such as "Catch-Up Strategy," government-funded incentive policies play a critical role in ensuring the synchronization of the quantity and quality of patents; moreover, from the perspective of this article, even for polluting firms, they can receive innovation incentives through the "leverage effect," then for non-polluting ones, such incentives should be even more efficacious.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

[1] China continues to be one of the most heavily polluted countries in terms of air quality, with its annual average PM2.5 concentration six times higher than the WHO guideline of 5 μg/m3. See "2024 Annual Update of Air Quality Life Index," Energy Policy Institute at the University of Chicago (EPIC). Available on: https://aqli.epic.uchicago.edu/files/AQLI-2024-Report_English.pdf. Since becoming the world's largest carbon emitter in 2014, the problem of CO_2 emissions has worsened, with China's greenhouse gas (GHG) emissions accounting for 25.88% of the global total. See "NDC Tracker," Climate Watch. Available on: https://www.climatewatchdata.org/2020-ndc-tracker.

[2] A commonly used indicator of patent quality is the citation counts. But citation counts suffer from difficulties similar to those of patent counts (Higham et al. 2021), especially because citation selection depends heavily on subjective judgment and the cognitive scope of inventors or examiners (Kelly et al. 2021).

[3] The reason is that when the firm's level of green R&D is still low (especially for the heavily polluting firms examined in this article), the R&D process itself involves a high entry threshold and requires substantial fixed-cost investment. As a result, even a small improvement in technological capability at this stage should correspond to a relatively large increase in cost. For the pollution governance process, we may likewise expect improvements in technological level to reduce pollution emissions. However, at this stage, pollution emissions should be less sensitive to technological improvement than costs are.

[4] To ensure the "comparability" of the individuals within the treatment and control group in the DID design (that is, as shown later in the article, the two groups should exhibit no significant pre-treatment differences), we performed sample matching. The matching variable was whether a firm was included in the CET pilot, and we used Caliper-Nearest matching to ensure that the matched sample could satisfy the parallel trend.

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Appendix A

Theoretical Model

Here, we present the technical details involved in the theoretical model in the main text that were omitted there due to space constraints.

We begin by returning to the convexity of the production function. We assume that all production functions are strictly convex, but the degree of convexity is not symmetric across different processes. In particular, we assume both the fixed-cost function associated with the green R&D process and the corresponding output function of the pollution abatement process are convex, but we further assume that the former is more strongly convex than the latter. It also implies a disproportionate relationship between cost and governance performance, namely, the improvement in pollution governance outcomes is slower than the increase in costs. Thus, $F'' < G'' = \frac{\partial^2 G}{\partial \gamma_g^2}$, as Figure A1 shows.

As the main text shows, put all terms together referred in the profit function of the firm, we have:

$$\begin{aligned}\Pi(\gamma_p, \gamma_g, N_p, N_g) \; &= p\gamma_p N_p^{\alpha} f - cN_g - F(\gamma_g) \\ &- \tau\left[\mathscr{R} + \left(\phi\gamma_p N_p^{\alpha} g\right)^{\eta} - \left(\phi\gamma_g N_g^{\beta} h\right)^{\theta}\right]\end{aligned} \tag{A1}$$

For investigating the optimal patent factor input involved in green R&D process, we can calculate the derivative of Π with respect to N_g and let it equal to 0:

$$\frac{\partial \Pi}{\partial N_g} = -c + \tau\gamma_g^{\theta}\beta\theta N_g^{\beta\theta - 1} = 0 \tag{A2}$$

Then the optimal N_g can be expressed as

$$N_g = \left[\frac{c}{\tau\theta\beta\gamma_g^{\theta}}\right]^{\frac{1}{\beta\theta - 1}} \tag{A3}$$

For investigating how environmental regulation affects green innovation quantity, we can further calculate the derivative of N_g with respect to τ:

$$\frac{\partial N_g}{\partial \tau} = -\frac{1}{\beta\theta - 1}\left[\frac{c}{\theta\beta\gamma_g^{\theta}}\right]\tau^{-\frac{1}{\beta\theta - 1} - 1} \tag{A4}$$

where $\beta\theta - 1 < 0$ for $0 < \theta, \beta < 1$. Then,

$$\frac{\partial N_g}{\partial \tau} > 0 \tag{A5}$$

meaning that, as environmental regulation becomes more stringent, the firm's optimal decision (holding other factors constant) is to choose a greater level of green innovation (patent) input in terms of quantity.

Now, consider the situation of corresponding patent factor input involved in the production process. Also, we calculate the derivative of Π with respect to N_p and let it be 0:

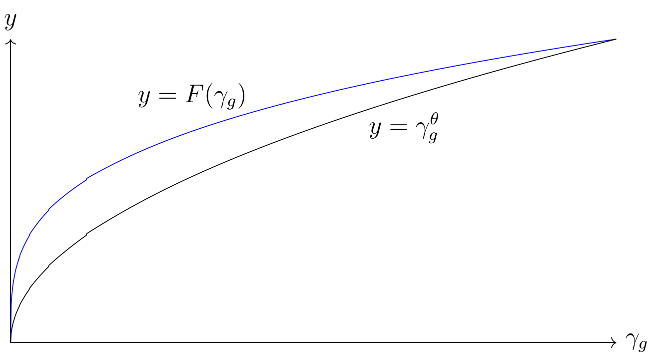

FIGURE A1 | The convexity of the fixed cost and pollution governance.

$$\frac{\partial \Pi}{\partial N_p} = p\gamma_p\alpha N_p^{\alpha - 1} - \tau\gamma_p^{\eta}\alpha\eta N_p^{\alpha\eta - 1} = 0 \tag{A6}$$

Then, by combining Equations (A2) and (A6), we can obtain:

$$\frac{c}{p\gamma_p\alpha N_p^{\alpha - 1}} = \frac{\gamma_g^{\theta}\beta\theta N_g^{\beta\theta - 1}}{\gamma_p^{\eta}\alpha\eta N_p^{\alpha\eta - 1}} \tag{A7}$$

By rearranging, the optimal N_g can be expressed as

$$N_p = \left[\frac{p\gamma_g^{\theta}\beta\theta}{\gamma_p^{\eta - 1}\eta}N_g^{\theta\beta - 1}\right]^{\frac{1}{\alpha(\eta - 1)}} \tag{A8}$$

To examine how green innovation input affects the non-green one in terms of quantity, namely, the innovation "leverage effect" or "crowding-out effect." We calculate the derivative of N_p with respect to N_g:

$$\frac{\partial N_p}{\partial N_g} = \frac{\theta\beta - 1}{\alpha(\eta - 1)}\left[\frac{p\gamma_g^{\theta}\beta\theta}{\gamma_p^{\eta - 1}\eta}\right]^{\frac{1}{\alpha(\eta - 1)}} N_g^{\frac{\theta\beta - 1}{\alpha(\eta - 1)} - 1} \tag{A9}$$

where $\frac{\theta\beta - 1}{\alpha(\eta - 1)} > 0$ for $0 < \theta, \beta, \alpha, \eta < 1$. Therefore,

$$\frac{\partial N_p}{\partial N_g} > 0 \tag{A10}$$

which means that, as green innovation-related patent input increases, non-green one also rises accordingly. Hence, green innovation exhibits a "leverage effect" on non-green innovation at the quantitative level.

Now, turn to investigate the optimal decision on green innovation quality the firm should choose. For this purpose, we calculate the derivative of Π with respect to γ_g, and let it equal to 0:

$$\frac{\partial \Pi}{\partial \gamma_g} = -F' + \tau\theta\gamma_g^{\theta - 1}N_g^{\theta\beta} = 0 \tag{A11}$$

By rearranging, we can obtain the following implicit function $\mathscr{J}$:

$$\mathscr{J}(\gamma_g, \tau) = \frac{F'}{\gamma_g^{\theta - 1}} - \tau\theta N_g^{\theta\beta} = 0 \tag{A12}$$

Taking the partial derivative of the above implicit function by applying the chain rule to, we can obtain the partial derivative of γ_g with respect to τ:

$$\left[\underbrace{F''\gamma_g}_{\text{negative}} - \underbrace{F'(\theta - 1)}_{\text{negative}}\right]\frac{1}{\gamma_g^{\theta}}\frac{\partial \gamma_g}{\partial \tau} = \theta N_g^{\theta\beta} \tag{A13}$$

where the first term within the bracket "[]" is negative for $F'' < 0$; and the second term is negative for $F' > 0$ and $0 < \theta < 1$. Thus, the sign of $\frac{\partial \gamma_g}{\partial \tau}$ is indeterminate. However, we will show that if F takes a specific convex functional form, for example, $F(\gamma_g) = \gamma_g^{\delta}$ with $0 < \delta < 1$, then the necessary and sufficient condition for $\frac{\partial \gamma_g}{\partial \tau} < 0$ is that the convexity of F be greater than that of the output function G for the pollution abatement process.

By casting $F = \gamma_g^{\delta}$, we can obtain the following first- and second-order derivatives of F:

$$\begin{cases} F' = \delta\gamma_g^{\delta - 1} \\ F'' = \delta(\delta - 1)\gamma_g^{\delta - 2} \end{cases} \tag{A14}$$

Then, the two terms within the bracket "[]" in Equation (A13) can be expressed as

$$F''\gamma_g - F'(\theta - 1) = \delta(\delta - \theta)\gamma_g^{\delta - 1} < 0 \tag{A15}$$

Here, $F''\gamma_g - F'(\theta - 1)$ will be totally negative for $\delta < \theta$, which means that

$$\frac{\partial \gamma_g}{\partial \tau} < 0 \tag{A16}$$

which implies that, as environmental regulation becomes more constringent, the firm will choose a lower level of innovation quality as its optimal decision. At the same time, the above deduction also implies that, as long as $\delta < \theta$ holds, the partial derivative of γ_g with respect to τ must be negative. In fact, In fact, when the quality of green innovation is at a relatively low level, $\delta < \theta$ is typically the necessary and sufficient condition for the convexity of F to be greater than the convexity of the γ_g-related term in G.

Last, we turn to examine how green innovation quality affects the non-green one, namely, the innovation "leverage effect" or "crowding-out effect" at the quality level. Also, we first calculate the derivative of Π with respect to γ_p and let it be 0:

$$\frac{\partial \Pi}{\partial \gamma_p} = pN_p^{\alpha} - \tau\eta\gamma_p^{\eta-1}N_p^{\alpha\eta} = 0 \tag{A17}$$

By combining Equations (A11) and (A17), we can obtain:

$$\frac{F'(\gamma_g)}{pN_p^{\alpha}} = \frac{\theta\gamma_g^{\theta-1}N_g^{\theta\beta}}{\eta\gamma_p^{\eta-1}N_p^{\alpha\eta}} \tag{A18}$$

By rearranging, we can obtain the expression of γ_p with respect to γ_g:

$$\gamma_p = \left[\frac{p\theta N_g^{\theta\beta}}{\eta N_p^{\alpha(\eta-1)}F'(\gamma_g)}\gamma_g^{\theta-1}\right]^{\frac{1}{\eta-1}} \tag{A19}$$

Then, calculate the derivative of γ_p with respect to γ_g, we have:

$$\frac{\partial \gamma_p}{\partial \gamma_g} = \frac{\theta-1}{\eta-1}\left[\frac{p\theta N_g^{\theta\beta}}{\eta N_p^{\alpha(\eta-1)}F'(\gamma_g)}\right]^{\frac{1}{\eta-1}}\gamma_g^{\frac{\theta-1}{\eta-1}-1} \tag{A20}$$

where $\frac{\theta-1}{\eta-1} > 0$ for $\theta, \eta < 1$, making

$$\frac{\partial \gamma_p}{\partial \gamma_g} > 0 \tag{A21}$$

which means that, as green innovation quality declines, non-green quality also declines accordingly. Hence, the quality of green innovation also exhibits some kind of "leverage effect" on non-green one.

Appendix B

Variable Specification

Here, we present the detailed specification of variables used in baseline regressions, which was not included in the main text due to space limitations.

Table B1 presents the name, abbreviation and measure methods of control variables. Table B2 presents the descriptive statistics of each variables.

TABLE B1 | Variable specifications.

Abbr.	Name	Description
asset	Firm asset	Logarithmic value of total firm asset recorded in the Balance Sheet
revenue	Operating revenue	Logarithmic revenue revealed by annual reports
age	Firm age	Logarithmic years from the date of establishment
flowratio	Flow ratio	The ratio of current assets to current liabilities
turnover	Turnover capability	Logarithmic days of turnover revealed by annual reports
DPS	Dividend per share	The ratio of total dividend to the nubmer of outstanding shares
incentive	Equity incentive	Whether there exists the equity incentive
lever	Leverage ratio	The ratio of equity capital to total asset
EPS_g	EPS growth	Annual growth rate of earning per share
ROE	Return on equity	The ratio of net profit to average stockholders' equity

TABLE B2 | Descriptive statistics of variables.

Statistic	*N*	Mean	St. dev.	Min	Max
GNinn	846	0.472	0.847	0.000	3.466
GNinv	846	0.338	0.701	0.000	3.045
GNuti	846	0.236	0.551	0.000	2.565
GNcom	846	0.168	0.323	0.000	0.997
regulation	846	0.002	0.002	0.000	0.006
asset	846	22.843	1.331	19.831	26.240
revenue	846	22.331	1.348	19.222	25.583
age	846	2.984	0.240	1.836	3.585
flowratio	846	1.675	1.579	0.219	12.925
turnover	846	4.613	0.827	2.721	6.904
DPS	846	0.174	0.233	0.000	1.500
incentive	846	0.176	0.381	0	1
lever	846	0.455	0.181	0.063	0.972
EPS_g	846	0.450	1.819	−25.077	9.240
ROE	846	0.099	0.083	−0.159	0.529